Engineering Mathematics - I · Unit_04 · Unit 4

DRAFT — UE26MA141A · Unit_04 — Unit 4 · Complete Unit Master

Notes-first study master · source-cited lecture explanations and compact page takeaways

Hugging Face PESU_Materials source tree, official units.json catalogue, transcript sidecars, slide text/PDF/PPT sources and the selected First_Year_8enterprise archive; see the provenance datasets in the release ZIP. Generated 18 October 2026.

· How to use this pack

Read the edited lesson notes by topic. Evidence chips point to transcript intervals and slide pages. The complete source transcripts, page text, formula candidates, and image locators are preserved in sibling JSON datasets. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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M1

Unit_04 · Unit 4

1 UE26MA141A · Unit_04: Unit 4

This unit master organizes source material for Engineering Mathematics - I, Unit_04, officially titled Unit 4. It includes 6 video transcripts and 11 source slide decks, with page-level text and rendered images where available. This draft preserves source wording and page order; it does not claim a complete human subject review or full audiovisual review. EV0000001

SHORT-NOTE RECAP

Unit map · UE26MA141A · Unit_04: Unit 4

Official catalogue entry: UE26MA141A, Unit_04 (Unit 4). Bucket folder present: yes. Videos: 6; slide decks: 11.

EV0000001

2 Gamma functions - definition, graph and properties

Study focus: Gamma functions - definition, graph and properties. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000002 · EV0000003

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

I welcome all of you to special functions, which is your fifth unit in engineering mathematics one.

Key details

- In this unit, you are going to learn gamma function, beta function, and basis function.
- In this particular class, we are going to discuss definition of gamma function.
- Properties of gamma function and graph of gamma function.
- Gamma function is a definite integral whose integrant depends on one variable.
- It is also known as Euler's integral of second kind.

- It is the generalization of factorial notation from integes to real numbers.
- We can define gamma function as.
- Gamma of n is equal to integral zero to infinity t power minus x power n minus one dx n greater than c.
- Thus gamma n defines a function of n for positive values of n.
- From this result we can obtain the value of gamma.
- Then we get gamma 1 is equal to integral 0 to infinity e power minus x power 1 minus 1 dx.
- x power 1 minus 1 is x power 0 and we know x power 0 is 1, so it just reduces to integral 0 to infinity e power minus x bx.
- And we know that integration of e power minus x dx is equal to e power minus x divided by minus 1.
- Using the fact that e power minus infinity is equal to 0 and e power 0 is equal to 1, we can evaluate this to 1.
- Thus we have gamma of 1 is equal to 1.
- We can have an alternate form of gamma function.
- We can obtain it easily by using definition of gamma function.
- We know that gamma n is equal to integral 0 to infinity t power minus x power n minus 1 dx.
- If x is t square, dx is 2t d.
- Therefore, gamma n is equal to integral 0 to infinity e power minus t square t square to the power n minus 1 2 t dt.
- Observe here, after this substitution, the limits of integration do not change because when x is equal to 0, your t is also 0.
- When x is infinity, t is also infinite.
- This is another form of gamma function.
- While solving problems on gamma function, we can make use of any of these two forms according to our convenience.
- Next, we will discuss some properties of gamma function.
- First of them is reduction formula.
- It states that gamma of n plus 1 is equal to n gamma n where n is a real number.
- We can prove this easily by using the definition of gamma function, which is.
- Gamma n is equal to integral 0 to infinity e power minus x power n minus 1 dx.
- We want gamma of n plus 1.
- which is equal to integral 0 to infinity e power minus x power n dx.
- We have to evaluate this integral by using integration by parts.

Example or application from the lecture

- In this expression, let us substitute x is equal to t square.
- We know already that integration by parts is given by integral uv dx is equal to u into integral v dx minus integral of derivative of u or d by dx of u into integral of v dx.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000002 · EV0000003 · EV0000004 · +1 source units in pack.json

Core explanation · 05:00–09:59

When you take x power n is equal to u and e power minus x is equal to v, you will get it as x power n e power minus x divided by minus 1 plus n into integral 0 to infinity.

Key details

- E power minus x power n minus one dx.
- When you substitute the limits of integration for the first term,.
- E power minus infinity zero and e power zero is one.
- So first term becomes a zero and observe the second term along with n, you have integral zero to infinity, e power minus x, x power n minus one.
- dx which is nothing but equal to gamma n.
- When n is a negative non-integer, the formula used to find gamma of n is gamma n is equal to gamma of n plus 1 divided by n.
- Remember, gamma n is not defined for n is equal to 0.
- And it is also not defined when n is a negative integer.
- But we can find the value of gamma n if n is negative real number which is not a negative integer using this formula.
- When n is a positive integer, gamma of n plus one is equal to n factory.
- We can prove this easily by using reduction formula which states gamma of n plus 1 is equal to n gamma n.
- That is 1, because we know gamma 1 is equal to 1.
- When we put n is equal to 2, we get gamma of 3, that is 2 gamma 2, which is 2 factorial.
- When you put n is equal to 3, we get gamma 4, which is 3 gamma 3, that is 3 factorial.
- And when you put n is equal to 4, we get gamma 5, which is 4 gamma 4, that is 4 factor.
- Just look at the pattern here.
- Gamma two is one factorial, gamma three is two factorial, gamma four is three factorial, and gamma five is four factorial.
- Gamma of n plus 1 is equal to n factorial where n is a positive integer.
- Remember, gamma n is not defined when n is 0 or when n is a negative integer.
- Gamma half value we can find now.
- Gamma half is equal to square root of pi.
- We can prove this by considering alternate form of gamma function, which is gamma n is equal to 2 times integral 0 to infinity e power minus x square x to the power 2n minus 1 dx.
- Which can also be written as two times integral 0 to infinity e power minus y square dy.
- Because since we are dealing with definite integrals, here x is a dummy variable.
- You can write it as x or y or t or any variable since we are dealing with definite integrals.
- which is equal to four times integral 0 to infinity 0 to infinity e power minus x square plus y square dx d y.
- Then x square plus y square is equal to r square.

Example or application from the lecture

- Let us put n is equal to 1 in this formula, we get gamma 2 is equal to 1, gamma 1.
- Let us put n is equal to half in this expression.

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EV0000006 · EV0000007 · EV0000008 · +1 source units in pack.json

Core explanation · 10:00–14:59

And dx dy is equal to r dr d theta.

Key details

- And observe the limits for x and y.
- Both are varying from 0 to infinity.
- That means both x and y are positive.
- So our region of integration is the first quadrant.
- In this first quadrant, r value varies from 0 to infinity and theta value varies from 0 to pi by 2.
- Then it can be written as four into integral 0 to pi by 2 b theta into integral 0 to infinity e power minus r square r br.
- When you integrate the first integral 0 to pi by 2 d theta, integration of d theta is theta, substitute the limits of integration, it reduces to 4 into pi by 2.
- into integral 0 to infinity e power minus r square r dr.
- That is equal to 2 by into minus 1 by 2 e power minus r square limits vary from 0 to infinity.
- Thus, we get gamma half the whole square is equal to pi, so we can conclude gamma half is equal to square root of pi, which is a very important result.
- This is the graph of gamma function.
- Observe here, gamma function is continuous for positive values of n and it is not continuous for negative values of n.
- We can find or you can plot this curve by using the properties gamma of n plus 1 is equal to n gamma n.
- Gamma n is equal to gamma n plus 1 divided by n.
- This is for negative real numbers which are not indigenous.
- And gamma half is equal to square root of pi.
- With these properties and the table, try to get the shape.
- This is very interesting thing.
- You can really do it.
- You will enjoy doing it.
- With this, we will end the session.

Example or application from the lecture

- Before that, just have a glance at this table where gamma n values are given for various values of n between 1 and 2.
- And also referring to the table which is given in the previous slide.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000010 · EV0000011 · EV0000012 · +1 source units in pack.json

3 Definition of Beta function and its properties

Study focus: Definition of Beta function and its properties. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000014 · EV0000015

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

In last class you have studied definition, properties and graph of gamma function.

Key details

- In this class, we are going to study definition of beta function and properties of beta function.
- Beta function is a definite integral whose integrant depends on two variables.
- It is also known as Euler's integral of first time.
- Beta function is defined as beta of m comma n is equal to integral 0 to 1 x power m minus 1 into 1 minus x power n minus 1 is, where m and n both are positive.
- Let me call this as equation number one.
- We can also have an alternate form for beta function.
- We can get it by putting x is equal to sine square theta in equation number 1.
- When x is equal to sine square theta, dx is equal to two sine theta cos theta d theta.
- Also, the limits of integration change with the substitution.
- When x is equal to 0, that implies sine square theta is equal to 0, that implies theta is equal to 0.
- When x is equal to 1, that means sine square theta is equal to 1, or theta is equal to pi by 2.
- With this, we will get beta of m comma n as integral 0 to pi by 2, sin square theta all power m minus 1.
- One minus sine square theta whole power n minus one into two sine theta cos theta b theta.
- 2 is a constant we can keep outside the integral sign.
- Sin square theta wall power m minus 1 is sin power 2 m minus 2 theta.
- One minus sin square theta we know it is cos square theta.
- Beta of m comma n is now two times integral 0 to pi by 2 sin power 2 m minus 1 theta cos power 2n minus 1 theta d theta.
- This is another form of beta function, we can also call it as trigonometric form of beta function.
- Both the forms are important, so you can remember both the forms here.
- Both are useful when solving the problems.
- We can choose any one of these forms according to our convenience.
- Next, we will study the properties of beta function.
- First property is symmetry, which states beta of m comma n is equal to beta of L comma n.
- We can prove this by using the definition of beta function, which states beta of m, n is equal to integral 0 to 1 x power m minus 1 into 1 minus x power n minus 1 dx.
- then dx becomes minus of dt.

- Also the limits of integration change with the substitution x equal to 1 minus t implies t is equal to 1 minus x.
- Therefore, when x is equal to 0, t is equal to 1.
- And when x is equal to 1, right, t is equal to 0.
- Therefore, now we can have beta of m comma n is equal to integral 1 to 0 1 minus t power m minus 1 t power n minus 1 minus t.
- T power n minus 1 minus t to the power m minus 2 t, which is nothing but equal to beta of n comma m.
- Thus we have proved now beta of m comma n is exactly same as beta of n comma m.
- And now we know beta of m n is same as beta of n m, so we can also write beta of m, comma n is equal to 2 times integral 0 to pi by 2 cos power 2 m minus 1 theta into sine power 2n minus 1 theta d.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000014 · EV0000015 · EV0000016 · +1 source units in pack.json

Core explanation · 05:00–09:59

To prove this, consider the diagnometric form of beta function that is beta of m, comma n is equal to 2 times integral 0 to pi by 2 sin power 2m minus 1 theta cos power 2n minus 1 theta d theta.

Key details

- Take 2m minus 1 is equal to p and 2n minus 1 is equal to q.
- Then we get m is equal to p plus 1 by 2 and n is equal to q plus 1 divided by 2.
- According to the definition of beta function, we should have m in beta of m comma n, m and n both are positive.
- So, m is greater than 0 implies now p plus 1 divided by 2 is greater than 0, that means p is greater than minus 1.
- This is what the conditions mentioned in property number 2.
- Therefore, integral 0 to pi by 2 sine power p theta cos power q theta d theta is equal to 1 by 2 times beta of p plus 1 by 2, comma q plus 1 divided by 2.
- This is also an important property.
- Whenever we come across the integrals in the form integral 0 to pi by 2, sin power p theta, cos power q theta, always we can express them in terms of beta functions.
- Already we proved now that integral 0 to pi by 2 sin power p theta cos power q theta d theta is 1 by 2 times beta of p plus 1 by 2 comma q plus 1 by 2.
- Then we get integral 0 to pi by 2 sin power p theta is e theta is equal to 1 by 2 times beta of p plus 1 by 2 comma 0 plus 1 divided by 2.
- Which is equal to 1 by 2 times beta of P plus 1 by 2, comma 1 by 2.
- And also we know that beta of m comma n is equal to beta of n, comma m.
- Thus we can conclude integral 0 to pi by 2 sine power p theta d theta is same as integral 0 to pi by 2 cos power p theta d theta and that is equal to half into beta of p plus 1 by 2 comma 1 by 2.
- Next property says that we can also express beta function in the form of an improper integral 0 to infinity x power m minus 1 divided by 1 plus x power m plus n dx.

- This property is also very useful because whenever we want to evaluate the integrals in the form integral 0 to infinity x power m minus 1 by 1 plus x power m plus n dx, always we can write them as beta of m comma n and we can find the value of beta of m comma n easily.
- Let me call this as equation number 1.

Example or application from the lecture

- In this let us put q is equal to 0.
- To prove this property, first let us consider the definition of beta function which says beta of m, n is equal to integral 0 to 1 x power m minus 1 minus x to the power n minus 1 bx.

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EV0000018 · EV0000019 · EV0000020 · +1 source units in pack.json

Core explanation · 10:00–14:59

That means dx is equal to minus 1 divided by 1 plus y the whole square dy.

Key details

- That implies y is equal to infinity.
- When x is equal to 1, 1 plus y is equal to 1, that means y is equal to 0.
- into 1 minus 1 by 1 plus y whole to the power n minus 1 into minus dy divided by 1 plus y the whole square, which is equal to minus of integral infinity to 0.
- 1 divided by 1 plus y whole power m minus 1 into 1 plus y minus 1 divided by 1 plus y whole to the power n minus 1.
- which is equal to integral 0 to infinity y power n minus 1 divided by 1 plus y to the power m minus 1 into 1 plus y to the power n minus 1 into 1 plus y whole square into dy.
- That is m plus n divide.
- Because since we are dealing with the definite integrals.
- X and Y both are dummy variables.
- We can use any variable here.
- Remember this, this is also an important result.
- This can be proved very easily by considering the definition of beta function.
- So first term becomes now integral 0 to 1 x power m minus 1 minus x to the power n plus 1 minus 1 dx.
- And second terms after applying bit of m per m formula, so we will get integral 0 to 1 x power m plus 1 minus 1 into 1 plus x to the power n minus 1 dx.
- That is equal to integral 0 to 1 x power m minus 1 minus x to the power n dx plus integral 0 to 1 x power m into 1 minus x to the power n minus 1 dx.
- 1 minus x power n minus 1 into x power m minus 1 is the common factor.

Example or application from the lecture

- Next consider the fourth property that says beta of m comma n can be written as beta of m comma n plus one plus beta of m plus one comma n.
- Let us use this definition.
- First let us consider the right hand side.

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EV0000022 · EV0000023 · EV0000024 · +1 source units in pack.json

Core explanation · 15:00–19:59

Take it outside.

Key details

- That is nothing but equal to beta of m comma n.
- Thus we have beta of m comma n is equal to beta of m comma n plus 1 plus beta of m plus 1 comma n.
- With this we will end this session.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000026 · EV0000027 · EV0000028 · +1 source units in pack.json

4 Relation between Beta and Gamma functions, Duplication formula

Study focus: Relation between Beta and Gamma functions, Duplication formula. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000030 · EV0000031

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

In last class you have learnt definition and properties of beta function.

Key details

- In this class, you are going to learn relationship between beta gamma function and duplication formula.
- It is a very important relation which is very useful and it states that beta of m comma n is equal to gamma of n into gamma of n divided by gamma of m plus n.
- integral 0 to infinity e power minus y square y power to n minus 1 dy.
- So here in gamma m I have used x and in gamma n I have used y doesn't matter because we are dealing with definite integrals.
- That is equal to 4 times integral 0 to infinity 0 to infinity e power minus x square into e power minus y square which is e power minus of x square plus y square into x power two m minus one into y power two n minus one dx dy.
- Therefore, your x square plus y square is now r square and dx dy is r into dr into d theta.
- In Cartesian form, we know that x varies from 0 to infinity and y also varies from 0 to infinity.

- That means x and y both are positive.
- So, region of integration is the first quadrant.
- R cos theta whole power two m minus one into r sin theta whole power two n minus one into r into dr into d theta.
- So first let me write all the terms containing r together and then write all the terms containing theta together.
- E power minus r square r power two m minus one r power two n minus one r into cos power two m minus one theta sin power two n minus one theta into d r d theta.
- Here you can observe, so e power minus r square into r power 2n minus 1 into r power 2n minus 1 into r is purely a function of r and cos power 2n minus 1 theta into sin power 2n minus 1 theta is purely a function of theta.
- Two times integral zero to infinity e power minus r square r power two m minus one r power two n minus one r into dr.
- You can observe here the first integral is nothing but equal to beta m comma n.

Example or application from the lecture

- First, let us see what is relationship between beta gamma functions.
- First, we will consider the definition of gamma function.
- Then consider gamma n.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000030 · EV0000031 · EV0000032 · +1 source units in pack.json

Core explanation · 05:00–09:59

e to the power minus t because we are putting r square is equal to t, so it is e power minus t into t to the power m minus one by two plus n minus one by two d t.

Key details

- We know that gamma n is equal to integral 0 to infinity e power minus t power n minus 1 dt.
- The expression whatever we got for gamma m into gamma n, which is beta of m, n into gamma m plus n.
- Thus we can conclude that beta of m comma n is equal to gamma m into gamma n divided by gamma of m plus n.
- It states that square root of pi into gamma of 2m is equal to 2 to the power 2m minus 1 into gamma of m into gamma of m plus 1 by 2.
- which is equal to two times integral 0 to pi by 2 sine power 2 n minus 1 theta cos power 2n minus 1 theta d theta.
- That means in the place of n I am putting m.
- Two variables there is only one variable.
- 2 m minus 1 theta d theta.
- That means 2 d theta is equal to dt.
- When theta is equal to 0, t is also equal to 0.
- When theta is equal to.
- Pi by two, your t is equal to pi.

- into dt by 2.
- And also you can use the result here 0 to pi sine power 2 m minus 1 t dt is equal to 2 times integral 0 to pi by 2 sin power 2 m minus 1 t.
- Thus my beta of m comma m now becomes two divided by two to the power two m minus one into integral zero to pi by two sine power two m minus one t dt, which is equal to one by two to the power two m minus one into beta of m comma half.
- That is by using the formula integral 0 to pi by 2 sine power p theta d theta is 1 by 2 into beta of p plus 1 by 2,1 by 2.
- We know that beta of m comma n is equal to gamma m into gamma n divided by gamma of m plus n.
- Apply this formula to the left hand side which is beta of m comma m so we get left hand side as gamma m into gamma m divided by gamma of m plus m which is gamma of two m.

Example or application from the lecture

- Next let us go for duplication formula for gamma function.
- To prove this, we first consider beta of m, n.
- we consider beta of m comma m.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000034 · EV0000035 · EV0000036 · +1 source units in pack.json

Core explanation · 10:00–14:59

Apply the same formula to the right hand side also, where 1 by 2 to the power 2m minus 1 is a constant and you have beta of m, half, so it is gamma half into gamma m divided by gamma of m plus half.

Key details

- We get gamma m divided by gamma 2m is equal to 1 by 2 to the power 2m minus 1 into square root of pi because we know gamma half is equal to square root of pi divided by gamma of m plus half.
- Thus we have square root of pi into gamma of 2m is equal to 2 to the power 2m minus 1 into gamma of m into gamma of m plus half.
- This is the required result.
- Since we need gamma n into gamma of 1 minus n,.
- And we know gamma 1 is equal to 1, so we have gamma n into gamma of 1 minus n is equal to beta of n, 1 minus n.
- And also we know beta of m comma n is equal to integral 0 to infinity x power n minus 1 divided by 1 plus x power m plus n dx.
- This is another form that is of a beta function.
- You have studied it.
- while studying the properties of beta function.
- Beta function as an improper integral.
- Now therefore we can have beta of n comma 1 minus n as.

- Integral zero to infinity x power n minus one divided by one plus x dx which is equal to pi divided by sin n pi.
- Why this is because of the fact that integral 0 to infinity x power n minus 1 divided by 1 plus x dx is equal to pi divided by sin n pi.
- The proof of this you will learn while studying the complex analysis.
- Thus, in this class you have studied relationship between beta gamma functions, duplication formula,.
- And also this result, which says gamma n into gamma of 1 minus n is equal to pi divided by sine n.
- With this we end this session.

Example or application from the lecture

- Next, we consider one important result that is gamma n into gamma of 1 minus n is equal to pi divided by sin n pi, where n lies between 0 and 1.
- To prove this, we consider the relationship between beta gamma function.
- Let us put M is equal to one minus N.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000038 · EV0000039 · EV0000040 · +1 source units in pack.json

5 Problems on Beta an Gamma functions

Study focus: Problems on Beta an Gamma functions. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000042 · EV0000043

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

In previous few classes, you have studied definition and properties of beta gamma functions, relationship between beta gamma function and duplication formula.

Key details

- In today's class, we are going to solve some problems on beta gamma function.
- Before that, we will recall some of the formulas you have studied earlier, which are useful in solving problems for today's class.
- First one is gamma n is equal to integral 0 to infinity e to the power minus x power n minus 1 dx.
- This is definition of gamma function.
- Second one is gamma of n plus 1 is equal to n gamma n.
- Third one is gamma half is equal to square root of pi.
- Porth one is beta of m comma n is equal to integral 0 to 1 x to the power m minus 1 into 1 minus x to the power n minus 1 dx.

- Next one is integral zero to pi by two sine power p theta cos power q theta t theta is equal to one by two times beta of p plus one by two comma q plus one by two, which is derived from the previous result itself.
- And the last one, but very important one, that is beta of m, comma n is equal to gamma m into gamma n divided by gamma of m plus n.
- That is the relationship between beta gamma function.
- Lear these formulas, all these formulas are very important.
- If minus log x is equal to t, x will be e to the power minus t, and dx is minus a power minus t dt.
- Therefore, when x is equal to 0, we have t is equal to log of 1 by 0.
- Thus you have when x is 0, t is equal to infinity, and when x is equal to 1, we have t is equal to log of 1 divided by 1, that is log 1 and log 1 is equal to 0.
- Therefore, integral 0 to 1 dx divided by square root of minus log x is equal to integral infinity to 0 minus a power minus d dt divided by square root of t.
- which is equal to integral 0 to infinity e to the power minus t.
- And minus of integral a to b is integral b to a so it is integral zero to infinity a power minus t to the power minus half dt.
- Integral zero to infinity e to the power minus t to the power minus one by two d we can write it as integral zero to infinity e power minus t to the power one by two minus one dt.
- And we have already studied the result that gamma half is equal to square root of pi.
- True that m plus 1 whole to the power n plus 1 into minus 1 power n into integral 0 to 1 x power m log x to the power n dx is equal to gamma of n plus 1.

Example or application from the lecture

- You can also consider the result gamma of n plus one is equal to n factorial when n is a positive integer which is derived from this result.
- Consider the first problem evaluate integral 0 to 1 dx divided by square root of minus log x.
- Which is log infinity, which you can consider it as infinite.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000042 · EV0000043 · EV0000044 · +1 source units in pack.json

Core explanation · 05:00–09:59

where n is a positive integer and m is greater than minus.

Key details

- In this integral, let me make some substitution so that I will get gamma function.
- In gamma function, we need an exponential term.
- When x is equal to e to the power minus y, dx is equal to minus of e to the power minus y, dy.
- log x is equal to minus y or we can say minus of log x is equal to y or we have log of 1 divided by x is equal to y which is.
- Similar to the previous problem, where also we got the same limits of integration.

- So here also when x is equal to 0 that means y is equal to log of 1 by 0 or log infinity which is infinity.
- When x is equal to 1, we have y is equal to log of 1 by 1 and log of 1 by 1 is log 1 and log one is equal to zero.
- log of e to the power minus y is.
- Minus y so it is minus y to the power n and in the last term we have already taken the negative sign outside so it is e to the power minus y d y.
- And now minus of integral infinity to 0, we can write it as integral 0 to infinity.
- And from this term, minus y whole to the power n, I have taken.
- Minus 1 to the power n outside.
- How we got this is.
- From the previous step, here you have e to the power minus m y, here you have e to the power minus y.
- Therefore, integral 0 to 1 x power m log x all power n dx is equal to minus 1 power n integral 0 to infinity e to the power minus u because we have taken m plus 1 y is equal to u into.
- Y power n y is u by m plus one.
- Evaluate integral 0 to 1 x to the power 4 1 minus x whole q dx.

Example or application from the lecture

- To prove this result, first let me consider the integral 0 to 1 x power m log x to the power n dx.
- Then m plus 1 into dy is equal to du and also you can consider here y is equal to u divided by m plus 1.
- Consider the left hand side that is integral 0 to 1 x power 4 1 minus x whole cube dx that can be written as x to the power 5 minus 1 into 1 minus x whole to the power 4 minus 1 dx.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000046 · EV0000047 · EV0000048 · +1 source units in pack.json

Core explanation · 10:00–14:59

Like this, just to compare this with the beta function definition, which is beta of m, comma n is equal to integral 0 to 1 x to the power m minus 1 into 1 minus x to the power n minus 1 dx.

Key details

- And now you can use the relationship between beta and gamma function which is beta of m comma n.
- If you want to write it in general, you can write it for beta of.
- M comma n which is equal to gamma of m into gamma of n divided by gamma of m plus n.
- Gamma four is three factorial and gamma nine is eight factorial.
- Where you have 1 minus x power n whole to the power r.
- But we want to write the integral on the left hand side in terms of beta function.
- And beta function definition says that beta of m comma n is equal to integral 0 to 1 x to the power m minus 1 into 1 minus x to the power n minus 20x.

- But in this problem we have x to the power n here.
- So to prove this result, first take x power n is equal to y.
- Then n into x power n minus 1 dx is equal to dy.
- With this substitution, the limits of integration will not change because when x is equal to 0, y is 0.
- When x is equal to 1, y is also equal to 1.
- What is dx? dx is equal to.
- From the substitution it is dy divided by n into x power n minus 1, which is nothing but n into y power n minus 1 divided by n.
- That is equal to 1 by n, we can keep it outside the integral sign, integral 0 to 1.
- and into 1 divided by y to the power n minus 1 divided by n.
- which is equal to y to the power m by n divided by y to the power 1 minus 1 by n, which is equal to y to the power m by n minus 1 plus 1 by n.
- So because now this integral, integral 0 to 1.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000050 · EV0000051 · EV0000052 · +1 source units in pack.json

Core explanation · 15:00–19:59

Y to the power m minus n plus one divided by n you can write it as m plus one divided by n minus one into one minus phi whole to the power r.

Key details

- R you can write it as r plus one minus one.
- Beta of m comma n.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000054 · EV0000055 · EV0000056 · +1 source units in pack.json

6 Problems on Beta and Gamma Function_continued

Study focus: Problems on Beta and Gamma Function_continued. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000058 · EV0000059

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

Consider the fifth problem evaluate integral zero to two pi sine power eight theta d theta.

Key details

- This I got by using one of the properties of the definite integral, which is integral 0 to 2a f of x dx is equal to 2 times integral 0 to a f of 2a minus x dx.

- If you use this property twice, you will get this.
- And after this, to simplify, I use one of the properties of beta function we have studied earlier, which is integral 0 to pi by 2, sine power p theta, cos power q theta.
- d theta is equal to one by two times beta of p plus one divided by two, comma q plus one divided by two.
- In this case, you have P is equal to 8 and Q is equal to 0.
- That is 8 plus 1 by 2, which is 9 by 2.
- Comma q plus one by two that is zero plus one divided by two that is one by two.
- And now use the property of gamma function that is gamma of n plus 1 is equal to n into gamma n.
- And gamma 7 by 2 can be written as gamma of 5 by 2 plus 1.
- Again gamma 5 by 2 can be written as 3 by 2 plus 1 and gamma 3 by 2 can be written as gamma of 1 by 2 plus 1.
- And we know gamma half.
- Gamma of 1 by 2 is equal to square root of pi.
- So thus we can write this as gamma half whole square in the numerator because there is one more gamma half and denominator gamma 5.
- Gamma 5 is equal to 4 factory.
- Next one is evaluate integral 0 to pi by 2 square root of cot theta d theta.
- cos theta divided by sine theta whole to the power one by two d theta, which is equal to integral zero to five by two sine power minus half theta cos power half theta d theta.
- Again it is in the form now integral 0 to pi by 2 sine power p theta cos power q theta d theta, which is equal to 1 by 2 times beta of p plus 1 divided by 2.
- comma q plus 1 divided by 2.
- In this particular problem we have p is equal to minus 1 by 2 and q is equal to 1 by 2.
- Because minus half plus 1 divided by 2 is 1 by 4, 1 by 2 plus 1 divided by 2 is 3 by 4.
- But how do we simplify gamma of 1 by 4 into gamma of t by 4?

Example or application from the lecture

- Consider integral zero to two pi sine power eight theta d theta, which is equal to four times integral zero to pi by two sine power eight theta d theta.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000058 · EV0000059 · EV0000060 · +1 source units in pack.json

Core explanation · 05:00–09:59

You have studied one result which says gamma of n into gamma of one minus n is equal to pi divided by sin n pi.

Key details

- You will get gamma of one by four into gamma of one minus one by four is equal to pi divided by sine of n into pi here n is equal to one by four, so it is sine of pi by four.
- When x square is tan theta, two x dx is equal to second square theta d theta.

- Therefore, integral zero to infinity dx divided by one plus x power four is equal to integral zero to pi by two one divided by one plus tan square because x square is tan.
- x power four is tan square theta into dx is second square theta d theta divided by two into square root of tan theta.
- Observe here the limits of integration change to 0 to pi by 2.
- How we got this is you have x square is equal to tan theta.
- So when x is equal to 0, that means tan theta is equal to 0.
- Tan theta is equal to zero implies theta is equal to zero.
- And when x is equal to infinity, that means tan theta is equal to infinity and we know tan theta is infinity when theta is equal to 5 bit.
- We know 1 plus tan square theta.
- 1 plus 10 square theta is equal to second square theta.
- Second to second square theta divided by two times ten theta.
- We can observe here we can cancel second square theta.
- Integral 0 to pi by 2 d theta divided by 2 into square root of tan theta.
- That is 1 by 2 into integral 0 to pi by 2.
- 1 by root tan theta we can write it as tan theta to the power minus half.
- into d theta.
- That is equal to 1 by 2 into integral 0 to pi by 2 sin theta divided by cos theta all to the power minus h into d theta.
- Which is equal to 1 by 2 into integral 0 to pi by 2, we get it as sin power minus 1 by 2 theta into cos power plus 1 by 2 theta.
- Cos power plus 1 by 2 theta d theta.
- So again it is equal to half into half into beta of three by four, comma, one by four because we know integral 0 to pi by 2 sin power p theta cos power q theta d theta.
- is equal to 1 by 2 times beta of p plus 1 divided by 2, comma q plus 1 divided by 2.
- We had p is equal to minus 1 by 2, q is equal to 1 by 2.
- It can be also written as beta of one by four comma three by four because we know that beta of m comma n is same as.

Example or application from the lecture

- Consider the seventh problem evaluate integral zero to infinity dx divided by one plus x to the power four.
- Term x square let us take it as tan theta.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000062 · EV0000063 · EV0000064 · +1 source units in pack.json

Core explanation · 10:00–14:59

beta of n comma n so we have already proved this property while studying the properties of beta function.

Key details

- Relationship between beta gamma function so it is one by four into gamma of three by four into gamma of one by four divided by gamma of one.
- Pi divided by 1 by square root of 2 which is equal to pi divided by root 2.
- Thus we have solved some problems on beta and gamma functions in this class.
- I hope you got an idea how to solve problems in this topic.
- Anyways, it is better to work on some more problems in this topic to get a better understanding.
- With this, we will end this session.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000066 · EV0000067 · EV0000068 · +1 source units in pack.json

7 Series solution of Besselâ??s differential equation, Bessel functions

Study focus: Series solution of Besselâ??s differential equation, Bessel functions. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000070 · EV0000071

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

We are discussing special functions.

Key details

- In today's class, you are going to learn Basel differential equation of order n, series solution of Basel differential equation, and Basel function.
- A linear second order ordinary differential equation in the form x square into y double dash plus x into y dash plus x square minus n square into y is equal to zero.
- Is known as Basel differential equation.
- Here, n is the order of the basal differential equation.
- Let me call this equal equation number 1.
- That is why we call it as series solution of the Basel differential equation.
- And article a0, a1, a2, etc. are constants and a0 is not equal to 0.
- So because we all know that d by dx of x to the power n is equal to n into x to the power n minus 1.
- Therefore, differentiation of x to the power r plus k is r plus k into x to the power r plus k minus 1.
- into x to the power r plus k minus 2.
- Here we have differentiated x to the power r plus k minus 1, which is r plus k minus 1 into x to the power r plus k minus 2.

- We get x square into summation r from 0 to infinity, ar into r plus k into r plus k minus 1.
- into x to the power r plus k minus 2 that is nothing but equal to y double dash this is y double dash then plus x into summation ar into.
- R plus k into x to the power r plus k minus 1.
- This is nothing but equal to y dash, then plus x square minus n square into summation ar into x to the power r plus k.
- This is nothing but equal to y.
- is equal to 0.
- Thus we have substituted the expressions for y dash and y double dash in the x equation x square y double dash plus x y dash plus x square minus n square into y is equal to z, which is a basal differential equation of order n.
- In the second term also I will take this x inside the summation.
- Thus, 1 is x square into summation ar x power r plus k.
- That is nothing but equal to summation r from 0 to infinity, ar into x to the power r plus k plus 2, then minus n square into summation r from 0 to infinity ar into x to the power r plus k is equal to 0.

Example or application from the lecture

- Let us call it as equation number 2, where k is a positive real number.
- Let us call it as equation number four.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000070 · EV0000071 · EV0000072 · +1 source units in pack.json

Core explanation · 05:00–09:59

That implies summation r from 0 to infinity ar into r plus k into r plus k minus 1 plus 1 into x to the power r plus k.

Key details

- We have considered the first two terms together here.
- Then minus n square into summation r from zero to infinity ar plus k plus summation r from zero to infinity ar x to the power r plus k plus two is equal to zero.
- Here we can cancel these two terms.
- Summation R from zero to infinity into x to the power r plus k plus two is equal to zero.
- Summation R from 0 to infinity R plus K whole square minus N square into AR into X to the power R plus K.
- Summation R from 0 to infinity is common.
- A R is common factor, x power r plus k is also common, so we can write it as r plus k whole square minus n square.
- Plus summation r from 0 to infinity a r into x to the power r plus k plus 2 is equal to 0.
- That implies summation r from 0 to infinity ar into x r plus k whole square minus n square into x to the power r plus k plus.
- Summation R is equal to 2 to infinity AR minus 2 into x to the power r plus k is equal to 0.
- See we obtain this term by replacing r by r minus 2 in the last term here.

- And here the summation starts from r minus 2 is equal to 0 to infinity, that is nothing but r is equal to 2 to infinity.
- In one summation R starts from 0 and in another R starts from 2.
- So, in the first summation, I will write two terms separately, that is, when r is equal to 0 and when r is equal to 1.
- That implies a0 into k square minus n square into x power k plus.
- A1 into K plus one whole square minus N square into X power K plus one plus summation R from two to infinity A R into R plus K whole square minus N square into X power R plus K plus.
- Summation R from 2 to infinity AR minus 2 into x power r plus k is equal to 0.
- After combining these two summations.
- We get it as summation r from two to infinity AR into R plus K whole square minus N square plus AR minus two whole into x power r plus k is equal to zero.
- Coefficient of x power k is a0 into k square minus n square.
- That implies k square minus n square is equal to 0 because we already know that k0 is not equal to 0.
- This implies a1 is equal to 0.
- So why we got this is because if a1 is not equal to 0, we get k plus 1 whole square minus n square is equal to 0.
- That means k plus 1 whole square is equal to n square.

Example or application from the lecture

- We consider it as a common summation.

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EV0000074 · EV0000075 · EV0000076 · +1 source units in pack.json

Core explanation · 10:00–14:59

That means k plus 1 is equal to plus or minus n, which is not correct because already we have proved now that k is equal to plus or minus n.

Key details

- Therefore, this term cannot be equal to 0.
- By equating the higher power of x to 0, we get ar is equal to some minus ar minus 2 divided by r plus k whole square minus n square.
- Call it as equation number five.
- First case is when k is equal to n.
- Put k is equal to n in equation number 5, we get ar is equal to minus ar minus 2 divided by r square plus 2 n r.
- Because we know that AR is equal to minus A R minus 2 divided by R plus K whole square minus n square by equation number 5.
- When you put k is equal to n in the denominator you get r plus n the whole square minus n square.
- And R plus N the whole square is R square plus 2 N R plus N square minus N square.

- Call it as equation number 6.
- A0 divided by 2 square plus 4n.
- Because when r is equal to 2, a r minus 2 becomes a 2 minus 2, which is a0.
- And in the denominator when you put r is equal to 2, r square becomes 2 square, 2n r becomes 4n.
- Which is equal to minus of a0 divided by 2 square into n plus 1.
- Therefore, your a2 is equal to minus a0 divided by 2 square into n plus 1.
- Call it as equation number 7.
- Now a3 is equal to minus a 1 divided by t square plus 6n, which is equal to 0 because a1 is equal to 0.
- This is got by ar is equal to minus ar minus 2 divided by r square plus 2 n r by putting r is equal to 3.
- In the numerator when R is 3, you get A3 minus 2, which is A1, which is there in the numerator here, and in the denominator 3 square you get and plus 2 Nr becomes a 6n.
- R is 4, so it becomes 4 square plus 2n r is equal to a n.
- Which is equal to minus 1 divided by 8 into m plus 2 into a2.
- We are substituting for a2 here because we know a2 is equal to minus a0 divided by 2 square into m plus 1.
- Thus we have a4 is equal to a0 divided by 32 into n plus 1 into n plus 2.
- which is equal to zero because we know that a3 is equal to zero.
- We can continue like this.
- Y is equal to a0 x power k plus a1 x power k plus one plus a2 x power k plus two plus a three x power k plus three plus a4 x power k plus four and so on.
- because we have assumed a y as a series solution that is y is equal to summation r from 0 to infinity ar into x to the power k plus r.
- We call it as Y1 because we are just discussing case 1 where your K is equal to plus N.
- We had two cases K is plus or minus N.
- So for the first case K is equal to N we call the solution as Y1.

Example or application from the lecture

- Let us substitute r is equal to 2, 3, 4, etc. in equation number 6, we get when you put r is equal to 2, we get a2, a2 is equal to minus.

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EV0000078 · EV0000079 · EV0000080 · +1 source units in pack.json

Core explanation · 15:00–19:59

So y1 is equal to a0 into x power n plus 0 into x power n plus 1 because we know that a1 is equal to 0 and k is equal to n here.

Key details

- Then plus a2 x power k plus 2.
- Next term is a3x power k plus 3.
- A3 is equal to 0.

- Then a4 x to the power k plus 4.
- Substitute for a4, a4 is a0 divided by 32 into n plus 1 into n plus 2 into x to the power n plus 4 plus other terms.
- That is y1 is equal to a naught into x power n minus 1 divided by 4 into n plus 1 into x power n plus 2 plus 1 divided by 32 into n plus 1 into m plus 2 into x to the power n plus 4 plus other terms.
- Remaining terms I write as it is.
- Minus 1 divided by 4 into n plus 1 into 2 power n into gamma of n plus 1 into x power n plus 2 plus.
- 1 by 32 into n plus 1 into n plus 2 into 2 power n into gamma of n plus 1 into x to the power n plus 4 plus other terms.
- Which is nothing but equal to gamma of n plus 2.
- And then you have n plus 2 into gamma of n plus 2.
- What is n plus 2 into gamma of n plus 2?
- So therefore y1 is equal to 1 divided by 2 power n into gamma of n plus 1 into x to the power n.
- Minus 1 divided by 2 square into 2 power n into gamma of n plus 2 into x power n plus 2.
- Plus 1 divided by 2 into 2 to the power 4.
- Because we had 32 in the denominator, 32 can be written as 2 to the power 5, it can be written as 2 into 2 to the power 4.
- into 2 power n into gamma of n plus 3 into x to the power n plus 4.
- Plus other terms.
- Plus this 1 by 2 we keep it as it is and 2 power 4 into 2 power n can be written as 2 to the power n plus 4 into gamma of n plus 3 into x to the power n plus 4 plus other terms.
- That is equal to C first term, so we can write it as minus 1 power 0.
- Minus 1 power 0 is 1.

Example or application from the lecture

- To standardize the result, let us take a0 is equal to 1 divided by 2 power n into gamma of n plus 1.
- And consider x power n divided by 2 power n, which can be written as x by 2 whole to the power n.

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EV0000082 · EV0000083 · EV0000084 · +1 source units in pack.json

Core explanation · 20:00–24:59

Then plus, since there is a negative sign here, we can write it as minus 1 whole to the power 1 divided by gamma of n plus 2 into x power n plus 2 divided by 2 to the power n plus 2, which is x by 2 whole to the power n plus 2.

Key details

- Next term is positive, so you can write it as 1 by 2 into minus 1 whole square in divided by gamma of n plus 3 into x power n plus 4 divided by 2 to the power n plus 4, which is x by 2

whole to the power n plus 4 plus other terms.

- That is equal to minus 1 power 0 divided by 0 factorial.
- 0 factorial is 1.
- Into gamma of n plus 1 into x by 2 whole to whole to the power n can be written as x by 2 whole to the power n plus 2 into 0.
- Plus minus 1 power n divided by 1 factorial, 1 factorial is 1, into gamma of n plus 2 into x by 2 whole to the power n plus 2 can be written as x by two whole to the power n plus 2 into 1.
- Plus, 1 by 2 can be written as 1 by 2 factorial and minus 1 whole square divided by gamma of n plus 3 into x by 2 whole to the power n plus 2 into 2.
- That is nothing but x by 2 whole to the power n plus 4 plus other terms.
- We can write it in the summation form as y1 is equal to summation r from 0 to infinity minus 1 to the power r into x by 2 whole to the power n plus 2 r into 1 divided by gamma of n plus r plus 1.
- into r factorial.
- This is because when r is equal to 0, you get minus 1 power 0 x by 2 whole to the power n plus 2 into 0 into 1 divided by gamma of n plus 1 into.
- One factorial, zero factorial.
- That is what the first term here.
- When you put r is equal to one, you get minus one to the power one into x by two whole to the power n plus two into one into one divided by gamma of n plus two into one factorial.
- When you put r is equal to two,.
- You get minus one whole square into x by two whole to the power n plus two into two into one divided by gamma of n plus two plus one that is gamma of n plus three into two factorial that is what the third term here.
- Call it as equation number nine.
- Equation number nine is called power series solution of basal differential equation.
- It is called Basel function of first order n.
- and it is denoted by jn of x.
- Thus jn of x is equal to summation r from 0 to infinity minus 1 power r divided by r factorial into gamma of n plus r plus 1 into x by 2 whole to the power n plus 2 r.
- Remember this expression, it's an important expression.
- Put k is equal to minus n in equation number five.
- Then we get j minus n of x is equal to summation r from 0 to infinity minus 1 power r x by 2 whole to the power minus n plus 2 r into 1 divided by gamma of minus n plus r plus 1 into r factorial.
- Basically, Paz's differential equation of order n is a differential equation of order 2.
- Moreover, we can.
- have a condition that n is a non integer here because we know that for a differential equation of order two, if y one and y two are two linearly independent solutions,.
- Then C1 by one plus c two by two that is the linear combination of those two linearly independent solutions is also a solution of the differential equation.
- Here jn and j minus c n are two linearly independent solutions only if n is not an integer.

Example or application from the lecture

- Thus the complete solution of Basel differential equation of order n is given by y is equal to c1 into j n of x plus c2 into j minus n of x, where c1 and c two are arbitrary constants.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000086 · EV0000087 · EV0000088 · +1 source units in pack.json

Core explanation · 25:00–29:59

If n is an integer, jn and j minus n are not linearly independent.

Key details

- You can prove this by taking the expression for jn of x.
- Just replace n by minus n.
- And hence you can prove that j minus n of x is minus 1 to the power n into jn of x.
- That shows that jn and j minus n are not linearly independent when n is an integer.
- Thus, if n is non-integer, then we have y is equal to c1 j n of x plus c2 j minus c n of x as the complete solution of the Bessel's differential equation.
- You can just try this to prove that jn and j minus n are linearly dependent when n is an integer.
- It is interesting, just take it as an assignment and try it on your own.
- With this, we will end this session.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000090 · EV0000091 · EV0000092 · +1 source units in pack.json

8 Source slide deck: introduction to Series solution

Study focus: Source slide deck: introduction to Series solution. The notes identify each page heading, summarize its main ideas, and explain its place in this unit; citations preserve the source page locator. EV0000307 · EV0000308

SHORT-NOTE RECAP

Study notes · Source slide deck: introduction to Series solution

The slide deck is rewritten here as searchable lesson notes. Each subsection states the slide heading and its concise teaching points; page citations lead back to the indexed source image when visual details are needed.

Slide 1 · [Image text, OCR]

- Department of Science and Humanities

Slide 2 · ENGINEERING MATHEMATICS I

- Department of Science and Humanities

Slide 3 · ENGINEERING MATHEMATICS - I

- Consider a homogeneous linear second order DE with variable co-efficients:
- Eq. (2) is called the normal form or canonical form or standard form of eq. (1).

Slide 4 · ENGINEERING MATHEMATICS - I

- A point = co is called an ordinary point of eq. (2), if P(c) and Q(c) are both analytic
- A point c = co is called a singular point of eq. (2) if either P(c) or Q(z) or both are
- A singular point is called a regular singular point of eq. (2) if (c - co)P(c) and (c

Slide 5 · ENGINEERING MATHEMATICS - I

- singular point is called a regular singular point of eq.
- 1. (2) if (x -co)P(α)
- I are not analytic at r = o.

Slide 6 · ENGINEERING MATHEMATICS - I

- Find the singular points and ordinary points of the equation
- ≥ 1 - α^2 = 0 → α2 = 1 → = ±1 are singular points of eq. (1)
- Dividing eq. (1) by (1 - r?) on both sides,

Slide 7 · ENGINEERING MATHEMATICS - I

- Comparing eq. (2) with the standard form i.e., y" + P(x)y' + Q(x)y = 0,
- → It is a regular singular point at c = 1.
- All the values of c ≠ ±l are the ordinary points of the equation.

Slide 8 · ENGINEERING MATHEMATICS - I

- about an ordinary point co be given as
- ≥ an(r - co)n be the series solution of the given eq.
- 5. Substitute @1, @2, ... in eq. (2) to obtain the required solution of the given equation.

Slide 9 · ENGINEERING MATHEMATICS - I

- Find the power series solution of the equation
- (x^2 + 1)y" + ry' - ry = 0 about r = 0.
- Let the series solution of eq. (1) be

Slide 10 · ENGINEERING MATHEMATICS - I

- Substituting y,y' and y" in eq. (1),
- To obtain a common power of c in each term, put n - 2 = m in the 1st term and n +1 = t
- Since m and t are dummy variables, replacing m and t by n:

Slide 11 · ENGINEERING MATHEMATICS - I

- Substituting y,y' and y" in eq. (1),

- To obtain a common power of c in each term, put n - 2 = m in the 1st term and n +1 = t
- Since m and t are dummy variables, replacing m and t by n:

Slide 12 · ENGINEERING MATHEMATICS - I

- → 2a2 + (6a3 + a1 -ao)c +
- ∑(n + 1)(n + 2)an+2 + n(n - 1)an + nan - an-1
- Equating the constant term, the coefficients of r and r^ to zero, we get:

Slide 13 · ENGINEERING MATHEMATICS - I

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 14 · [Image text, OCR]

- Department of Science and Humanities

EV0000307 · EV0000308 · EV0000309 · +39 source units in pack.json

9 Source slide deck: Power series Continued

Study focus: Source slide deck: Power series Continued. The notes identify each page heading, summarize its main ideas, and explain its place in this unit; citations preserve the source page locator. EV0000349 · EV0000350

SHORT-NOTE RECAP

Study notes · Source slide deck: Power series Continued

The slide deck is rewritten here as searchable lesson notes. Each subsection states the slide heading and its concise teaching points; page citations lead back to the indexed source image when visual details are needed.

Slide 1 · [Image text, OCR]

- Department of Science and Humanities

Slide 2 · ENGINEERING MATHEMATICS I

- SUB-TOPIC: POWER SERIES SOLUTION_continued
- Department of Science and Humanities

Slide 3 · ENGINEERING MATHEMATICS - I

- Find the power series solution about the point co = 2 of the equation
- We solve the differential equation about the ordinary point To = 2:
- Step 1: Assume a power series about r = 2

Slide 4 · ENGINEERING MATHEMATICS - I

- nan(x - 2)n-1 = (n + 1)an+1(x - 2)n,
- n(n - 1)an(z - 2)n-2 =
- (c - 2)y' = ∑nan(x - 2)n =

Slide 5 · ENGINEERING MATHEMATICS - I

- Step 2: Substitute into the differential equation
- (n+ 2)(n + 1)an+2(c - 2)n +
- nan(c - 2)n + ≥(n + 1)an+1(x - 2)n +

Slide 6 · ENGINEERING MATHEMATICS - I

- (n + 2)(n + 1)an+2 + (n + 1)an+1 + (n + 1)an = 0.
- Step 4: Compute the coefficients

Slide 7 · ENGINEERING MATHEMATICS - I

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 8 · [Image text, OCR]

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EV0000349 · EV0000350 · EV0000351 · +21 source units in pack.json

10 Source slide deck: Frobenius Method

Study focus: Source slide deck: Frobenius Method. The notes identify each page heading, summarize its main ideas, and explain its place in this unit; citations preserve the source page locator. EV0000373 · EV0000374

SHORT-NOTE RECAP

Study notes · Source slide deck: Frobenius Method

The slide deck is rewritten here as searchable lesson notes. Each subsection states the slide heading and its concise teaching points; page citations lead back to the indexed source image when visual details are needed.

Slide 1 · [Image text, OCR]

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Slide 2 · ENGINEERING MATHEMATICS I

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Slide 3 · ENGINEERING MATHEMATICS - I

- Series solution when c = O is a regular singular point of the equation
- 1. Assume the solution of eq. (1) to be
- equation (known as the Indicial Equation).

Slide 4 · ENGINEERING MATHEMATICS - I

- Case I: When roots are distinct and do not differ by an integer, then the complete
- Case III: When roots are distinct and differ by an integer, making a co-efficient of y
- series become infinite when m = m1, we modify the form of y by replacing ao by bo(m - m1).

Slide 5 · ENGINEERING MATHEMATICS - I

- Solve in series the equation
- Solution. Given eq.: 9c(1 - c)y" - 12y' + 4y = 0
- Put r = 0, Po(O) = 0 = = 0 is a singular point.

Slide 6 · ENGINEERING MATHEMATICS - I

- y" = m(m - 1)aoxm-2 + (m + 1)mazm-1 + (m + 2)(m + 1)a2xm + ..
- 9z(1 - x)|m(m - 1)aoxm-2 + (m + 1)ma1cm-1 + (m + 2)(m + 1)a2αm
- 9m(m - 1)aozm-1 + 9(m + 1)ma1α"m + 9(m + 2)(m + 1)a2xm+1 + ..

Slide 7 · ENGINEERING MATHEMATICS - I

- The lowest power of r is m - 1. So equating its coefficient to zero (coefficient of ao):

- 9m(m - 1)ao - 12mao = 0 → ao [9m² - 9m - 12m] = 0 ≥ ao [9m² - 21m] = 0
- i.e., roots are distinct but do not differ by an integer.

Slide 8 · ENGINEERING MATHEMATICS - I

- 9(m + 2)(m + 1)a2 - 9(m + 1)ma1 - 12(m + 2)a2 + 4a = 0
- → a2 [9(m + 2)(m + 1) - 12(m + 2)] = a1(9m² + 9m - 4)
- → (m + 2)a2[9m + 9 - 12] = a1(9m² - 3m + 12m - 4)

Slide 9 · ENGINEERING MATHEMATICS - I

- 32(m + 2)(m + 1)
- . y1 = (y)m=0 = ao + a1x + a2x² + ..· = ao +

Slide 10 · ENGINEERING MATHEMATICS - I

- (y)m=7/3 = α7/3

Slide 11 · ENGINEERING MATHEMATICS - I

- Hence, the complete solution is
- y = C1(y)m + C2(y)m2

Slide 12 · [Image text, OCR]

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EV0000373 · EV0000374 · EV0000375 · +33 source units in pack.json

11 Source slide deck: Frobenius method Continued

Study focus: Source slide deck: Frobenius method Continued. The notes identify each page heading, summarize its main ideas, and explain its place in this unit; citations preserve the source page locator. EV0000409 · EV0000410

SHORT-NOTE RECAP

Study notes · Source slide deck: Frobenius method Continued

The slide deck is rewritten here as searchable lesson notes. Each subsection states the slide heading and its concise teaching points; page citations lead back to the indexed source image when visual details are needed.

Slide 1 · [Image text, OCR]

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Slide 2 · ENGINEERING MATHEMATICS I

- SUB-TOPIC: frobenius method_continued

- Department of Science and Humanities

Slide 3 · ENGINEERING MATHEMATICS - I

- Find the general solution of 8r2y" + 10ry' - (1 + c)y = 0.
- We solve the differential equation using the Frobenius method about the regular singular
- Step 1: Assume a Frobenius series

Slide 4 · ENGINEERING MATHEMATICS - I

- Step 2: Substitute into the differential equation
- $8x^2y'' + 10y' -(1 + c)y = 0$,
- (8(n + r)(n + r - 1) + 10(n + r) - 1)an - an-1

Slide 5 · ENGINEERING MATHEMATICS - I

- The coefficient of r" gives
- 8r(r - 1) + 10r - 1 = 0,

Slide 6 · ENGINEERING MATHEMATICS - I

- 8(n + r)(n + r - 1) + 10(n + r) - 1
- First solution (r = 1/4). The denominator simplifies to

Slide 7 · ENGINEERING MATHEMATICS - I

- The first few coefficients are
- Second solution (r = -1/2). The denominator becomes
- $8n^2$ - 6n = 2n(4n - 3).

Slide 8 · ENGINEERING MATHEMATICS - I

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 9 · [Image text, OCR]

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EV0000409 · EV0000410 · EV0000411 · +24 source units in pack.json

12 Source slide deck: Bessel functions of order n and n

Study focus: Source slide deck: Bessel functions of order n and n. The notes identify each page heading, summarize its main ideas, and explain its place in this unit; citations preserve the source page locator. EV0000487 · EV0000488

SHORT-NOTE RECAP

Study notes · Source slide deck: Bessel functions of order n and n

The slide deck is rewritten here as searchable lesson notes. Each subsection states the slide heading and its concise teaching points; page citations lead back to the indexed source image when visual details are needed.

Slide 1 · [Image text, OCR]

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Slide 2 · ENGINEERING MATHEMATICS I

- SUB-TOPIC: BESSEL FUNCTIONS of order n and -n
- Department of Science and Humanities

Slide 3 · ENGINEERING MATHEMATICS - I

- Bessel functions of order n and -n

Slide 4 · ENGINEERING MATHEMATICS - I

- Bessel functions of order n and -n

Slide 5 · ENGINEERING MATHEMATICS - I

- Bessel functions of order n and -n

Slide 6 · ENGINEERING MATHEMATICS - I

- Bessel functions of order n and -n

Slide 7 · [Image text, OCR]

- Department of Science and Humanities

EV0000487 · EV0000488 · EV0000489 · +18 source units in pack.json

13 Historical question-source pages · unit mapping candidates

Study focus: Historical question-source pages · unit mapping candidates. The notes identify each page heading, summarize its main ideas, and explain its place in this unit; citations preserve the source page locator. EV0000508 · EV0000509

SHORT-NOTE RECAP

Study notes · Historical question-source pages · unit mapping candidates

The slide deck is rewritten here as searchable lesson notes. Each subsection states the slide heading and its concise teaching points; page citations lead back to the indexed source image when visual details are needed.

Archive page 2 from Series Solution of Bessel__1GK7DgclHD.pdf; unit match candidate score 0.5833; source text/OCR may be partial. Class content ☒Series Solution of Bessel's Differential Equation ENGINEERING MATHEMATICS I ☒Bessel differential equation of order n ☒Bessel Function

Archive page 4 from Series Solution of Bessel__1GK7DgclHD.pdf; unit match candidate score 0.4167; source text/OCR may be partial. Series Solution of Bessel Differential Equation 0 t,) e 2 L (r k r r y a x ☒ + = = −−−−− ☒ 0 where, is a positive real number ' are constants and k 0 ra a s ☒ ( )( ) 2 2 2 0 1 (4 an) d r k r r d y a r k r k x dx ☒ + − = = + + − −−−−− ☒ ( ) 1 0 T en, (3 h) r k r r dy a r k x dx ☒ + − = = + −−−−− ☒ ENGINEERING MATHEMATICS I

Archive page 5 from Series Solution of Bessel__1GK7DgclHD.pdf; unit match candidate score 0.4167; source text/OCR may be partial. Series Solution of Bessel Differential Equation Substituting equation and in equati (2), (3) (4) (1) on we get, ( )( ) 2 2 0 1 r k r r x a r k r k x ☒ + − = + + − ☒ + ( ) 1 0 r k r r x a r k x ☒ + − = + ☒ + ( ) 2 2 0 r k r r x n a x ☒ + = − ☒ ☒ ( )( ) 0 1 r k r r a r k r k x ☒ + = + + − ☒ + ( ) 0 r k r r a r k x ☒ + = + ☒ + 2 0 r k r r a x ☒ + + =☒ ENGINEERING MATHEMATICS I 0 = 2 0 r k r r n a x ☒ + = −☒ 0 =

Archive page 6 from Series Solution of Bessel__1GK7DgclHD.pdf; unit match candidate score 0.4167; source text/OCR may be partial. Series Solution of Bessel Differential Equation ☒ ( )☒ ☒ 0 1 1 r k r r a r k r k x ☒ + = + + −+ ☒ + 2 0 0 r k r r a x ☒ + + = = ☒ ENGINEERING MATHEMATICS I 2 0 r k r r n a x ☒ + = −☒ ☒ ( ) 2 0 r k r r a r k x ☒ + = + ☒ 2 0 r k r r n a x ☒ + = −☒ + 2 0 0 r k r r a x ☒ + + = = ☒ ☒ ( ) ☒ ☒ 2 2 0 r k r r a r k n x ☒ + = + − ☒ + 2 0 0 r k r r a x ☒ + + = = ☒ ☒ ( ) ☒ ☒ 2 2 0 r k r r a r k n x ☒ + = + − ☒ + 2 2 0 r k r r a x ☒ + − = = ☒

Archive page 7 from Series Solution of Bessel__1GK7DgclHD.pdf; unit match candidate score 0.4167; source text/OCR may be partial. Series Solution of Bessel Differential Equation ENGINEERING MATHEMATICS I ☒ ☒ ☒ 2 2 0 k a k n x − + ( ) ☒ ☒ 2 2 1 1 1 k a k n x + + − ( ) ☒ ☒ 2 2 2 r k r r a r k n x ☒ + = + − ☒ + + 2 2 0 r k r r a x ☒ + − = = ☒ ☒ ☒ ☒ 2 2 0 k a k n x − + ( ) ☒ ☒ 2 2 1 1 1 k a k n x + + − ( ) ☒ ☒ 2 2 2 2 r k r r r a r k n a x ☒ + − = ☒ ☒ + − + ☒ ☒ ☒ + 0 = Equating the co-efficients of to zero, we , get kx 2 2 0( ) 0 a k n − = 2 2 0 k n ☒ − = ☒ ( ) 0 0 a ☒ k n = ☒ 1 Equating the co-efficient s of to zero, kx + we get, ( ) ( ) 2 2 1 1 0 a k n + − = ☒ By equating the higher x powers of to zero, we get, ( ) 2 2 2 r r a a r k n − = − + − 1 0 a = (5) −−−−−−−− for 2,3,4,5..... r =

Archive page 8 from Series Solution of Bessel__1GK7DgclHD.pdf; unit match candidate score 0.4167; source text/OCR may be partial. Series Solution of Bessel Differential Equation ENGINEERING MATHEMATICS I Let us consider the two cases: case ( ) : i k n = Put in equation 5 ( ) k n = Then we get, ( ) 2 2 2 r r a a r k n − ☒ ☒ = − ☒ ☒ ☒ ☒ + − ☒ ☒ (6) −−−−−−− 2,3,4,..... Put in (6) we . , . get r = 0 2 2 2 4 a a n = − + (7) −−−−−−−− ( ) 0 2 2 1 a n = − + ( ) 0 2 2 1 a n = − + 2 a ☒ 2 2 2 r r a a r nr − = − +

Archive page 9 from Series Solution of Bessel__1GK7DgclHD.pdf; unit match candidate score 0.4167; source text/OCR may be partial. Series Solution of Bessel Differential Equation ENGINEERING MATHEMATICS I 2 4 2 4 8 a a n = − + 1 3 2 3 6 a a n = − + 0 = ( ) 1 0 a = 2 2 we have 2 , r r a a r nr − ☒ ☒ = − ☒ ☒ + ☒ ☒ 0 4 32(1)(2) a a n n ☒ = + + ( ) 0 2 1 8 2 2 (1) a n n ☒ ☒ = − − ☒ ☒ + + ☒ ☒ (8) −−−−−−−− 0 2 2 2 (1) a a n ☒ ☒ = − ☒ ☒ + ☒ ☒

Archive page 10 from Series Solution of Bessel__1GK7DgclHD.pdf; unit match candidate score 0.4167; source text/OCR may be partial. Series Solution of Bessel Differential Equation 3 5 2 5 10 a a n = − + ENGINEERING MATHEMATICS I 0 = . . . ' substituting s (2), in weget, ra 1 2 3 4 0 1 2 3 4 ..... k k k k k y a x a x a x a x a x + + + + = + + + + We have, by (2), 1 0 n y a x = 3 0 n x + + ☒ 1 0 n x + + ☒ ( ) 2 0 4 1 n a x n + − + 4 0 32(1)(2) n a x n n + + + + ... +

Archive page 11 from Series Solution of Bessel__1GK7DgclHD.pdf; unit match candidate score 0.4167; source text/OCR may be partial. Series Solution of Bessel Differential Equation ENGINEERING MATHEMATICS I i.e. ( ) 2 4 1 0 1 1 ..... 4 1 32(1)(2) n n n y a x x x n n n + + ☒ ☒ ☒ ☒ = − + + ☒ ☒ + + + ☒ ☒ ☒ ☒ To standardize the result ,let us take, 0 1 2 (1) n a n = ☒ + then we get , ( ) 2 4 1 1 1 1 ..... 2 (1) 4 1 32(1)(2) n n n n y x x x n n n n + + ☒ ☒ ☒ ☒ = − + + ☒ ☒ ☒ + + + + ☒ ☒ ☒ ☒ ( ) 2 4 1 1 1 ..... 2 (1) 4 1 2 (1) 32(1)(2)2 (1) n n n n n n x x x n n n n n n + + = − + + ☒ + + ☒ + + + ☒ +

Archive page 12 from Series Solution of Bessel__1GK7DgclHD.pdf; unit match candidate score 0.4167; source text/OCR may be partial. Series Solution of Bessel Differential Equation ( ) ( ) 2 1 0 1 1 2 1 ! n r r r x y n r r + ☒ = ☒ ☒ = − ☒ ☒ ☒ + + ☒ ☒ ☒ 1 1 2 (1) n n y x n = ☒ + ENGINEERING MATHEMATICS I 2 4 2 4 1 1 1 1 .... 2 (1) 2 (2) 2 2 (3) n n n n n n n x x x n n n + + + + = − + + ☒ + ☒ + ☒ + ( ) 0 1 (1) 2 n x n − ☒ ☒ = ☒ ☒ ☒ + ☒ ☒ ( ) 0 2 0 1 0! (1) 2 n x n + ☒ − ☒ ☒ = ☒ ☒ ☒ + ☒ ☒ (9) −−−−−− 2 2 1 2 2 (2) n n x n + − ☒ ☒ + 4 4 1 2 2 2 (3) n n x n + + ☒ ☒ ☒ + ... + ( ) 1 2 1 (2) 2 n x n + − ☒ ☒ + ☒ ☒ ☒ + ☒ ☒ ( ) 2 4 1 1 2 (3) 2 n x n + − ☒ ☒ + ☒ ☒ ☒ + ☒ ☒ .... + ( ) 1 2 1 1 1! (2) 2 n x n + ☒ − ☒ ☒ + ☒ ☒ ☒ + ☒ ☒ ( ) 2 2 2 1 1 2! (3) 2 n x n + ☒ − ☒ ☒ + ☒ ☒ ☒ + ☒ ☒ .... +

Archive page 13 from Series Solution of Bessel__1GK7DgclHD.pdf; unit match candidate score 0.6667; source text/OCR may be partial. Series Solution of Bessel Differential Equation Thus, ☒It is called Bessel function of first kind of order n. ☒Equation (9) Is called power series solution of Bessel differential equation. ☒It is denoted by ( ) n J x ( ) 2 0 1 ( ) ! (1) 2 r n r n r x J x r n r + ☒ = − ☒ ☒ = ☒ ☒ ☒ + + ☒ ☒ ☒ ENGINEERING MATHEMATICS I

Archive page 14 from Series Solution of Bessel__1GK7DgclHD.pdf; unit match candidate score 0.4167; source text/OCR may be partial. Series Solution of Bessel Differential Equation case ( ) : ii k n = − Put in equatio n (5) k n = − ( ) ( ) 2 0 1 (!) 1 2 1 n r r n r x J x n r r −+ ☒ − = ☒ ☒ = − ☒ ☒ ☒−+ + ☒ ☒ ☒ Then we get, ENGINEERING MATHEMATICS I

Archive page 15 from Series Solution of Bessel__1GK7DgclHD.pdf; unit match candidate score 0.4167; source text/OCR may be partial. Thus the complete solution of Bessel differential equation is given by 1 2 ( ) ( ) n n y C J x C J x − = + provided n is a non integer. ENGINEERING MATHEMATICS I Series Solution of Bessel Differential Equation ( ) 1 2 and are arbitraryconstants C C

EV0000508 · EV0000509 · EV0000510 · +10 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Official unit catalogue	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/units.json	Official course/unit title and index.
Gamma functions - definition, graph and properties transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/AV_Summaries/001_Gamma_functions_-_definition_graph_and_properties__c183e86bfc3ac0a2.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Gamma functions - definition, graph and properties transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/AV_Summaries/001_Gamma_functions_-_definition_graph_and_properties__c183e86bfc3ac0a2.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Gamma functions - definition, graph and properties transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/AV_Summaries/001_Gamma_functions_-_definition_graph_and_properties__c183e86bfc3ac0a2.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9982; only the JSON segmentation supplies the displayed timestamps.
Gamma functions - definition, graph and properties transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/AV_Summaries/001_Gamma_functions_-_definition_graph_and_properties__c183e86bfc3ac0a2.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Definition of Beta function and its properties transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/AV_Summaries/002_Definition_of_Beta_function_and_its_properties__c1ec05b61754baa1.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Definition of Beta function and its properties transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/AV_Summaries/002_Definition_of_Beta_function_and_its_properties__c1ec05b61754baa1.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Definition of Beta function and its properties transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/AV_Summaries/002_Definition_of_Beta_function_and_its_properties__c1ec05b61754baa1.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9985; only the JSON segmentation supplies the displayed timestamps.
Definition of Beta function and its properties transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/AV_Summaries/002_Definition_of_Beta_function_and_its_properties__c1ec05b61754baa1.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Relation between Beta and Gamma functions, Duplication formula transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/AV_Summaries/003_Relation_between_Beta_and_Gamma_functions_Duplication_formula__982d3dde7e270fc7.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Relation between Beta and Gamma functions, Duplication formula transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/AV_Summaries/003_Relation_between_Beta_and_Gamma_functions_Duplication_formula__982d3dde7e270fc7.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Relation between Beta and Gamma functions, Duplication formula transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/AV_Summaries/003_Relation_between_Beta_and_Gamma_functions_Duplication_formula__982d3dde7e270fc7.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9980; only the JSON segmentation supplies the displayed timestamps.
Relation between Beta and Gamma functions, Duplication formula transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/AV_Summaries/003_Relation_between_Beta_and_Gamma_functions_Duplication_formula__982d3dde7e270fc7.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Problems on Beta an Gamma functions transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/AV_Summaries/004_Problems_on_Beta_an_Gamma_functions__ccd478a3bed2d97a.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Problems on Beta an Gamma functions transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/AV_Summaries/004_Problems_on_Beta_an_Gamma_functions__ccd478a3bed2d97a.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Problems on Beta an Gamma functions transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/AV_Summaries/004_Problems_on_Beta_an_Gamma_functions__ccd478a3bed2d97a.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9987; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Problems on Beta an Gamma functions transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/AV_Summaries/004_Problems_on_Beta_an_Gamma_functions__ccd478a3bed2d97a.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Problems on Beta and Gamma Function_continued transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/AV_Summaries/005_Problems_on_Beta_and_Gamma_Function_continued__5f7e0c86ddf2398a.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Problems on Beta and Gamma Function_continued transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/AV_Summaries/005_Problems_on_Beta_and_Gamma_Function_continued__5f7e0c86ddf2398a.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Problems on Beta and Gamma Function_continued transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/AV_Summaries/005_Problems_on_Beta_and_Gamma_Function_continued__5f7e0c86ddf2398a.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9982; only the JSON segmentation supplies the displayed timestamps.
Problems on Beta and Gamma Function_continued transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/AV_Summaries/005_Problems_on_Beta_and_Gamma_Function_continued__5f7e0c86ddf2398a.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Series solution of Besselâ??s differential equation, Bessel functions transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/AV_Summaries/010_Series_solution_of_Besselâ_s_differential_equation_Bessel_functions__0b6c9f8e6bffd10d.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Series solution of Besselâ??s differential equation, Bessel functions transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/AV_Summaries/010_Series_solution_of_Besselâ_s_differential_equation_Bessel_functions__0b6c9f8e6bffd10d.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Series solution of Besselâ??s differential equation, Bessel functions transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/AV_Summaries/010_Series_solution_of_Besselâ_s_differential_equation_Bessel_functions__0b6c9f8e6bffd10d.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9989; only the JSON segmentation supplies the displayed timestamps.
Series solution of Besselâ??s differential equation, Bessel functions transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/AV_Summaries/010_Series_solution_of_Besselâ_s_differential_equation_Bessel_functions__0b6c9f8e6bffd10d.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Slide page text: Gamma Function	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/001_Gamma_Function__00935a1aa845f4cb.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Gamma Function	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/001_Gamma_Function__00935a1aa845f4cb.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.

Source	Path	Use
Page-render source for Gamma Function (converted PDF; original hash fb0f3bbeb4bc9e91a815c62874ef14de7f58b751a004ad6c54b652bccfab9429)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/001_Gamma_Function__00935a1aa845f4cb.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Beta Function	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/002_Beta_Function__a7f5e3feeb91e390.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Beta Function	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/002_Beta_Function__a7f5e3feeb91e390.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Beta Function (converted PDF; original hash 9d6a1d1f962fe15e6c957ef3379b68b52d9d2a8aa9803bdc9d95ac791fef02dd)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/002_Beta_Function__a7f5e3feeb91e390.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Relation between beta and gamma functions	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/003_Relation_between_beta_and_gamma_functions__4b16e06ca7aa6dc4.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Relation between beta and gamma functions	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/003_Relation_between_beta_and_gamma_functions__4b16e06ca7aa6dc4.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.

Source	Path	Use
Page-render source for Relation between beta and gamma functions (converted PDF; original hash 11134e6e9a904408513e6f892e1b5a00a4af0ddd2603e89ad1e2912a6055ad27)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/003_Relation_between_beta_and_gamma_functions__4b16e06ca7aa6dc4.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Problems on Beta and Gamma Functions	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/004_Problems_on_Beta_and_Gamma_Functions__cef7f8e988425437.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Problems on Beta and Gamma Functions	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/004_Problems_on_Beta_and_Gamma_Functions__cef7f8e988425437.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Problems on Beta and Gamma Functions (converted PDF; original hash 75162fc20072e4d69545d66f5bc6ddc7262b679c4cddd51270d7e317f40f1b81)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/004_Problems_on_Beta_and_Gamma_Functions__cef7f8e988425437.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Problems on Beta and Gamma Functions Continued	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/005_Problems_on_Beta_and_Gamma_Functions_Continued__b61a87e2ba951d7e.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Problems on Beta and Gamma Functions Continued	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/005_Problems_on_Beta_and_Gamma_Functions_Continued__b61a87e2ba951d7e.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.

Source	Path	Use
Page-render source for Problems on Beta and Gamma Functions Continued (converted PDF; original hash ccf0075124a4efd6b73292a676cf072c47bf12005afe28a69b35c36f8a30baa2)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/005_Problems_on_Beta_and_Gamma_Functions_Continued__b61a87e2ba951d7e.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: introduction to Series solution	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/006_introduction_to_Series_solution__273a7378aa2cf38d.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: introduction to Series solution	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/006_introduction_to_Series_solution__273a7378aa2cf38d.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for introduction to Series solution (converted PDF; original hash bb598f809b2c77ec29e57e49ea30074f6b2d59877db910c8835dddd52ec124ab)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/006_introduction_to_Series_solution__273a7378aa2cf38d.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Power series Continued	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/007_Power_series_Continued__ccfc5caf892a036c.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Power series Continued	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/007_Power_series_Continued__ccfc5caf892a036c.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.

Source	Path	Use
Page-render source for Power series Continued (converted PDF; original hash af23c8f1ad5d56206fe5c548a8779ae8daf6acf2554a18d766e8a359588d9772)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/007_Power_series_Continued__ccfc5caf892a036c.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Frobenius Method	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/008_Frobenius_Method__e4086c57d59143d5.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Frobenius Method	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/008_Frobenius_Method__e4086c57d59143d5.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Frobenius Method (converted PDF; original hash d59bdb13f4eaaaeb79bf4f66bf3429b0091d457a548b6aa4396550239ca8365a)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/008_Frobenius_Method__e4086c57d59143d5.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Frobenius method Continued	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/009_Frobenius_method_Continued__5bafed2d527576e2.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Frobenius method Continued	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/009_Frobenius_method_Continued__5bafed2d527576e2.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.

Source	Path	Use
Page-render source for Frobenius method Continued (converted PDF; original hash 8f64f3331f5de673a7f2e637c4c81f4b3b6bcec79f1343e30794d0bbafbd866e)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/009_Frobenius_method_Continued__5bafed2d527576e2.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Series solution of Bessel differential equation	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/010_Series_solution_of_Bessel_differential_equation__182c999b377f4fc6.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Series solution of Bessel differential equation	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/010_Series_solution_of_Bessel_differential_equation__182c999b377f4fc6.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Series solution of Bessel differential equation (converted PDF; original hash 76969bce20da67eb569a4e523bf3d9be3e0ec9c57606847ce60e42285d3178b0)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/010_Series_solution_of_Bessel_differential_equation__182c999b377f4fc6.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Bessel functions of order n and n	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/011_Bessel_functions_of_order_n_and_-n__4d450476e26e4e4f.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Bessel functions of order n and n	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/011_Bessel_functions_of_order_n_and_-n__4d450476e26e4e4f.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.

Source	Path	Use
Page-render source for Bessel functions of order n and n (converted PDF; original hash 8908775f3114e13979f0e25f05c9130615d83c425c6155065017c2168b4a4929)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/Slides/011_Bessel_functions_of_order_n_and_-n__4d450476e26e4e4f.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Extracted historical pages: Series Solution of Bessel__1GK7DgclHD.pdf	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/First_Year_8enterprise/Files/Math 1/Extra/Extra Questions/Series Solution of Bessel__1GK7DgclHD.pdf	Candidate-matched page text excerpt; original PDF bytes, hash and path remain in datasets/question-archive.json.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
2. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
3. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
4. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00001	1 covered · 0 excluded · 0 gaps	catalogue: Unit title and index copied from the official units.json catalogue; human content review is pending.

Source	Unit counts	Example decisions
SRC00002	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0052. Sidecar variants compare...
SRC00003	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0052. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00004	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0052. Sidecar variants compare...
SRC00005	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0052. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00006	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0059. Sidecar variants compare...
SRC00007	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0059. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00008	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0059. Sidecar variants compare...
SRC00009	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0059. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00010	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0055. Sidecar variants compare...
SRC00011	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0055. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00012	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0055. Sidecar variants compare...
SRC00013	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0055. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00014	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0059. Sidecar variants compare...
SRC00015	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0059. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00016	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0059. Sidecar variants compare...
SRC00017	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0059. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00018	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0044. Sidecar variants compare...
SRC00019	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0044. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00020	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0044. Sidecar variants compare...
SRC00021	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0044. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00022	6 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0060. Sidecar variants compare...
SRC00023	6 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0060. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00024	6 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0060. Sidecar variants compare...
SRC00025	6 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0060. Sidecar variants compare...
SRC00026	0 covered · 0 excluded · 16 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...

Source	Unit counts	Example decisions
SRC00028	0 covered · 0 excluded · 16 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00027	0 covered · 0 excluded · 16 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00029	0 covered · 0 excluded · 16 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00031	0 covered · 0 excluded · 16 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...

Source	Unit counts	Example decisions
SRC00030	0 covered · 0 excluded · 16 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00032	0 covered · 0 excluded · 11 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00034	0 covered · 0 excluded · 11 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00033	0 covered · 0 excluded · 11 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...

Source	Unit counts	Example decisions
SRC00035	0 covered · 0 excluded · 22 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00037	0 covered · 0 excluded · 22 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00036	0 covered · 0 excluded · 22 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00038	0 covered · 0 excluded · 6 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00040	0 covered · 0 excluded · 6 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...

Source	Unit counts	Example decisions
SRC00039	0 covered · 0 excluded · 6 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00041	14 covered · 0 excluded · 0 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00043	14 covered · 0 excluded · 0 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00042	14 covered · 0 excluded · 0 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...

Source	Unit counts	Example decisions
SRC00044	8 covered · 0 excluded · 0 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00046	8 covered · 0 excluded · 0 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00045	8 covered · 0 excluded · 0 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00047	12 covered · 0 excluded · 0 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00049	12 covered · 0 excluded · 0 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...

Source	Unit counts	Example decisions
SRC00048	12 covered · 0 excluded · 0 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00050	9 covered · 0 excluded · 0 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00052	9 covered · 0 excluded · 0 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00051	9 covered · 0 excluded · 0 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...

Source	Unit counts	Example decisions
SRC00053	0 covered · 0 excluded · 17 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00055	0 covered · 0 excluded · 17 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00054	0 covered · 0 excluded · 17 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00056	7 covered · 0 excluded · 0 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00058	7 covered · 0 excluded · 0 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...

Source	Unit counts	Example decisions
SRC00057	7 covered · 0 excluded · 0 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00059	13 covered · 0 excluded · 0 gaps	q2: Original archive PDF page 2; candidate unit association is text-based and unconfirmed. q4: Original archive PDF page 4; candidate unit association is text-based and unconfirmed. q5: Original archive PDF page 5; candidate unit association is text-based and unconfirmed.

C Revision and retrieval practice